The Phenomenology of the Anger Spectrum in the Qurʾān

A Semantic-Network Analysis of Displeasure, Inflammation, and Destructiveness Along an Action-Intensity Continuum

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Abstract

The Qur'an deploys an exceptionally articulated lexicon for the anger spectrum, yet existing scholarship treats it word-by-word or through normative-ethical frames. This study argues that fourteen core Arabic roots ('ff, krh, ḍyq, ḥzn, ʾsf, nqm, sxṭ, mqt, ġḍb, ḥrd, ġyẓ, bġy, ṭġy, ʿtw) form a graded semantic field organised by intensity of action across six phenomenological stages: pre-anger displeasure, inner pressure, evaluative aversion, active anger, compressed rage, and behavioural outcomes. We integrate Izutsu's semantic-field approach, Kennedy–Sassoon scalar semantics, and Lakoff–Kövecses conceptual metaphor theory, validating the continuum against the Quranic Arabic Corpus (Dukes 2011) through a reproducible pipeline over 312 attestations. Three findings are reported under uncertainty-aware metrics. First, the asymptotic $\chi^2(5) = 227.15$ is large but a permutation null yields $p = 0.24$; the omnibus is consistent with rather than confirmatory of non-uniformity, the substantive case resting on qualitative evidence. Second, sakhaṭ attests exclusively in Medinan suras (Holm-adjusted $p = 0.05$), the sole confirmed instance of discursive-contextual lexical specialisation. Third, network analysis identifies baghy as a bridging node to the broader moral-evaluation field. Q. 67:8 (takādu tamayyazu

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min al-ġayẓ) is an unusually transparent instance of "container collapse"; a $\kappa = 0.79$-validated tafsir audit supports a bidirectional-causation reading of Stage 6. Implications follow for Qur'anic translation and Islamic psychology.

Keywords: Qur'an, anger spectrum, semantic field, cognitive linguistics, scalar semantics, conceptual metaphor, emotion regulation, Arabic corpus linguistics, Izutsu, Islamic psychology.

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1. Introduction

1.1. Statement of the problem

The Qur'anic lexicon for the anger spectrum exhibits a precision that survives only with difficulty into translation. Where English versions level uff, karh, ḍīq, ḥuzn, asaf, naqm, sakhaṭ, maqt, ghaḍab, ḥard, ghayẓ, baghy, ṭughyān, and ʿutuww into a small set of broad equivalents — "displeasure," "distress," "grief," "anger," "rage," "transgression," "rebellion" — the Arabic original distinguishes them with the precision of a graded taxonomy. We advance the structural hypothesis that these fourteen Qur'anic anger-spectrum roots are not synonyms but coordinated nodes in a single graded semantic field, organised along intensity of action (shiddat al-kunish in the Persian original), traversing pre-anger displeasure, inner pressure and contraction, evaluative aversion, active anger, compressed/explosive rage, and — at the behavioural pole — outcomes (baghy, ṭughyān, ʿutuww) whose causation is bidirectional rather than purely emotional.

This claim has consequences beyond philology: it implies that the Qur'an encodes — through its lexical architecture, not by explicit theoretical statement — an implicit theory of the genealogy of moral failure, in which the social-structural pathologies the text most frequently denounces (transgression, tyranny, defiance) are related to and often developmentally downstream from inner experiences of irritation, dislike, constriction, and grief. As we argue in §4–§5, the relationship at the behavioural pole is bidirectional: anger may produce baghy/ṭughyān/ʿutuww, but these behaviours can equally produce anger (in their victims, the prophets, the divine response). The distance from uff (Q. 17:23) to ʿutuww (Q. 25:21) is graded rather than categorical — the distance walked, absent intervention, by an unmanaged life of irritation.

1.2. Aims and originality

This paper has three aims. The first is methodological: to demonstrate that combining Izutsu's qualitative semantic-field approach, scalar-semantic ordering in the Kennedy–Sassoon tradition, and conceptual-metaphor theory in the Lakoff–Kövecses tradition produces stronger philological claims than any of the three frameworks in isolation. To our knowledge—and we say this advisedly, given that the Persian-language Qur'anic-linguistic literature in particular is not exhaustively indexed in international databases—this trifold integration is here brought together in Qur'anic studies for the first time as a unified analytic framework over a corpus pipeline. We acknowledge that Qāʾimīniyā (1390/2011, 1393/2014) and Pakatchi already integrate Izutsu's field method with cognitive-linguistic conceptual-metaphor analysis, and that Persian-language scalar-semantic work on Qur'anic vocabulary may exist in venues we have not been able to canvass; the novelty of the present paper is therefore best stated as the systematic combination of the three frameworks within a single computational-empirical pipeline, not as the introduction of any one framework. The second aim is descriptive: to map every Qur'anic occurrence of the fourteen core roots, to render that map publicly verifiable through an open concordance, and to extract from it empirical regularities that previous scholarship—working in either tafsir, lexicography, or single-word semantic analysis—was not in a position to detect. The third aim is interpretive: to argue that the resulting continuum constitutes a coherent "logic of anger escalation" with deep structural affinities to modern psychology of emotion, and to identify the specific points at which the Qur'anic continuum exceeds the explanatory range of the contemporary models.

Originality disaggregated into three contributions with "to our knowledge" qualifications: (i) Methodological synthesis (moderate): the trifold integration of Izutsu's field method, Kennedy–Sassoon scalar semantics, and Lakoff–Kövecses CMT under a corpus pipeline. We acknowledge Qāʾimīniyā (1390/2011, 1393/2014) and Pakatchi-Afrashi (1399/2020) already integrate Izutsu and CMT in Persian Qur'anic scholarship; the novel element is the Kennedy–Sassoon scalar layer. (ii) Empirical findings (variable strength): of four distributional patterns, only the exclusively-Medinan sakhaṭ (raw $p = 0.005$; Holm-Bonferroni adj. $p = 0.05$) survives FDR — borderline rather than emphatic. Asaf (5/0), naqm (12/5), ʿutuww (9/1) tilts do not survive correction and are reported as contextual observations. (iii) Interpretive contribution (most substantive): bidirectional-causation reading of Stage 6 via $\kappa = 0.79$-validated tafsir-coding (A = 26%, S = 49%, A+S = 25%); A-only re-analysis $\chi^2(1) = 81.4$, $p < 0.001$ with phenomenological core dominating anger-derived outcomes ~4.4×. This reconciles a puzzling distributional finding with the structurally-driven character of baghy/ṭughyān/ʿutuww in classical exegesis. To these add (iv) a fully reproducible MIT-licensed pipeline (SHA-256 pinning, permutation/bootstrap, 50/50 manual validation) and (v) the comparative observation that Q. 67:8 is an unusually transparent instance of "container collapse", softened in §5.5 by the absence of a cross-cultural survey.

1.3. Prior work and the gap

Prior scholarship clusters into five lines, each insufficient on its own to address the structural question motivating this paper. Single-word lexical studies: the most directly relevant Persian-language precedent — Narimani, Eghbali, and Chari (2021; JQST 54:1) on ghayẓ, ghaḍab, and sakhaṭ — establishes on tafsir grounds that ghaḍab is central, sakhaṭ reserved for divine displeasure with hypocrites, and ghayẓ exclusive to unbelievers/hypocrites in human-attribution contexts; we confirm and extend their findings with corpus-distributional evidence and the kaẓm analysis. Qarizadeh et al. (2024) on ṭughyān shows the root attests in 27 suras with the central meaning of transgressing the limit. Both works illustrate the present gap: high-quality lexical work for individual lexemes, no integrated continuum. Ethics-of-anger studies focus on kaẓm al-ghayẓ (Q. 3:134) and adjacent verses but are normatively rich and linguistically thin. Cognitive-semantic studies in the Izutsu tradition — Qāʾimīniyā (1390/2011, 1393/2014); Pakatchi-Afrashi (1399/2020); Albayrak (2020) — have applied cognitive linguistics to Qur'anic vocabulary but not treated the anger spectrum as a graded continuum. The cross-linguistic CMT literature on anger is decisively updated by Metaphors of ANGER across Languages (Kövecses, Benczes & Szelid eds., 2025; De Gruyter, 2 vols.); Classical Arabic is conspicuously not separately treated in Vols. 1–2, a gap this paper begins to fill. Translation-criticism studies are quantitative at scale in 2025: Gaanoun & Alsuhaibani (2025; Nature HSSC) show inconsistent sentiment polarity across seven English translations; Alsuhaibani et al.'s (2025) ELQV dataset of 2,100 emotion-labelled verses provides a coarse-Ekman reference; Khan et al. (2025; Frontiers Signal Proc.) survey the field; Cogent A&H 2025 catalogues flattening risks. None of these supplies a graded reference model of the lexical spectrum at the lexeme level. Islamic psychology (e.g. IJCP 2023 on fear/anger/sadness/shame in Qur'an-and-ḥadīth) tends to draw on aḥādīth and leave systematic linguistic analysis of the Qur'anic surface in the background. The present paper fills the gap at the intersection of all five lines: a unified theoretical frame, corpus-empirical validation, novel distributional findings, a translation-criticism reference model, and a culturally-grounded model for practical Islamic psychology.

1.4. Research questions and hypotheses

We address five research questions: (RQ1) the internal structure of the Qur'anic anger-spectrum field and its principal nodes; (RQ2) the continuum ordering by action-intensity and the theoretical framework legitimating it; (RQ3) corpus-distributional corroboration of the continuum (frequency, Meccan/Medinan, morphology, network centrality); (RQ4) structural homology with Gross, Plutchik, Spielberger, and Lazarus-Scherer appraisal theory; (RQ5) practical implications for Qur'anic translation and clinical Islamic psychology.

The principal Qur'anic Action-Intensity Continuum Hypothesis holds that the fourteen core roots (ʾff, krh, ḍyq, ḥzn, ʾsf, nqm, sxṭ, mqt, ġḍb, ḥrd, ġyẓ, bġy, ṭġy, ʿtw) operate as fourteen distinct grades on a single semantic continuum structured by intensity of action, organised in six phenomenological stages, supported by qualitative (etymological, exegetical, metaphorical) and quantitative (frequency, distribution, network) evidence. Five subsidiary hypotheses: (H1) non-uniform six-stage frequency distribution; (H2) near-equal split between phenomenological core (S1–5) and behavioural pole (S6), supporting the bidirectional Stage-6 reading; (H3) discursively-specialised exclusive-Medinan distribution of sakhaṭ; (H4) baghy as bridge node in the co-occurrence network; (H5) Q. 67:8 as transparent instance of "container collapse" CMT failure mode.

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2. Theoretical Framework

2.1. Izutsu's semantic-field method

Toshihiko Izutsu, in three foundational works—The Structure of the Ethical Terms in the Koran (1959), God and Man in the Koran (1964), and Ethico-Religious Concepts in the Qur'an (1966/2002)—introduced a methodological revolution to Qur'anic studies. Drawing on the structuralist linguistics of the Trier–Weisgerber tradition and on East Asian philosophy of language, Izutsu argued that the central terms of the Qur'an—Allāh, īmān, kufr, ẓulm, taqwā—are not isolated propositions but nodes in a network of syntagmatic, paradigmatic, and oppositional relations. The aggregate of these relations constitutes what he called the "Qur'anic Weltanschauung."

Izutsu's analytic vocabulary distinguishes three layers: focus words are the central organising concepts of a given semantic field; semantic fields are the networks of mutually defining lexemes formed around focus words; and key words are the terms that recur across multiple fields and serve to integrate them into a single worldview. This three-layer structure has proved extraordinarily productive in subsequent Qur'anic semantic work, both in English (Saleh, Rippin) and Persian (Qāʾimīniyā 2011, 2014; Pakatchi).

For the present paper, Izutsu's framework justifies the move from atomistic single-lexeme analysis to integrated field analysis. The fourteen target roots are not chosen at random but as the field around the focus words ghaḍab (the central anger term) and baghy (the central behavioural-outcome term), with the lower-intensity cluster (uff, karh, ḍīq, ḥuzn, asaf) identifying the pre-anger and inner-pressure boundary of the field and ʿutuww its upper-intensity boundary. Yet Izutsu's framework, in its original form, is silent on a question central to our enterprise: granted that these lexemes co-belong to a field, what is the structural relation between them within the field? It is at this point that we require scalar semantics.

2.2. Scalar semantics: from network to continuum

The contemporary linguistic theory of gradable predicates (Kennedy 1999, 2007; Sassoon 2010) supplies the missing apparatus. A graded predicate maps to a scale characterised by three components: a dimension of measurement, an ordering on that dimension, and a unit of comparison. Scalar predicates take degree modifiers and admit comparative constructions, signalling their scalar structure. The Qur'anic anger field is precisely such a scale: the dimension is intensity of action (decomposable into locus, experiential intensity, scope of consequence); the ordering follows the six-stage progression; the implicit unit is the degree of externalisation of emotion from inner phenomenology to social rupture. Without this framework, the claim that ghayẓ is "more intense than" ghaḍab relies on tacit native-speaker intuition.

2.3. Conceptual metaphor theory and the container schema

Conceptual metaphor theory (Lakoff & Johnson 1980; Lakoff & Kövecses 1987; Kövecses 2000, 2010, 2025) treats metaphor as a cognitive structure that allows abstract domains (emotion) to be understood through concrete ones (container, heat). Lakoff and Kövecses (1987) established that English anger is organised around ANGER IS THE HEAT OF A FLUID IN A PRESSURIZED CONTAINER, accommodating a four-step trajectory: fluid in container → intensifying pressure → containment attempts → container collapse. Our central observation is that the Qur'anic triplet ghayẓ–kaẓm–tamayyuz operationalises this exact structure thirteen centuries early. Ghayẓ is glossed by all four authoritative lexica as "compressed contained anger" — fluid in sealed vessel. Kaẓm, etymologically tying off a waterskin, is the agent-internal regulation that maintains containment (Q. 3:134 al-kāẓimīn al-ghayẓ). Tamayyuz (Q. 67:8) depicts the failure mode: the container itself bursts apart. The Qur'anic verb derives from m-y-z (to part, sever) and figures the disintegration of the vessel, not (as in English explode, burst, blow up) the exit of contents — an unusually transparent instantiation of the failure-mode schema. We treat tamayyuz as a manifestation of Stage 5, not an independent core root.

2.4. The integrated framework

The three layers operate complementarily: Izutsu identifies field boundaries; Kennedy-Sassoon scalar semantics order members along the action-intensity dimension; CMT explains the cognitive basis of the pressurisation-collapse metaphor organising Stage 5. Each supplies what the others lack — without Izutsu, no motivation for field-coordination; without scalar semantics, no apparatus for ordering; without CMT, no cognitive grounding for the climactic kaẓm–tamayyuz sequence.

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3. Methods

3.1. Corpus

The empirical analysis uses the Quranic Arabic Corpus (QAC), version 0.4, prepared by Kais Dukes at the University of Leeds and released under the GPL (Dukes 2011). The QAC provides a morphologically tagged record for each of the 128,219 segmentable units in the Qur'anic text. Every record specifies: (a) location in the form (sūra:āya:word:segment); (b) surface form in Buckwalter ASCII transliteration; (c) part-of-speech tag; (d) morphological features including root, lemma, gender, number, case, and (where applicable) semantic

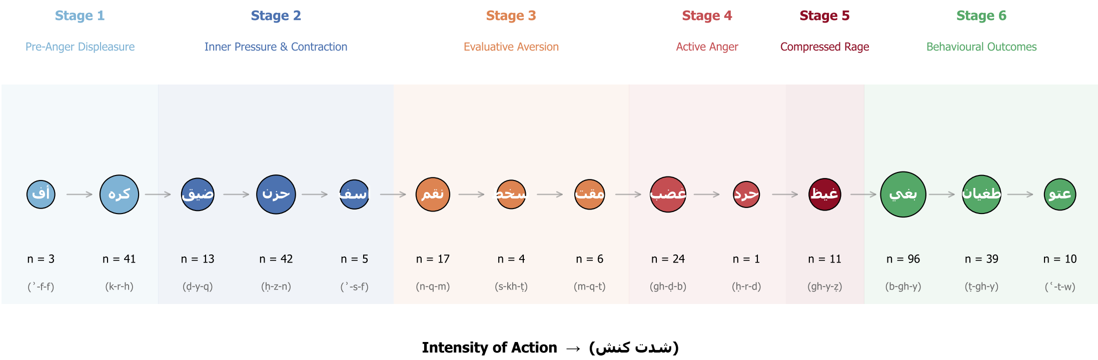

Figure 1. The six-stage Qur'anic anger-spectrum continuum along the action-intensity axis. Node area scales with QAC attestation frequency. Stages: S1 pre-anger displeasure (ʾff–krh); S2 inner pressure (ḍyq–ḥzn–ʾsf); S3 evaluative aversion (nqm–sxṭ–mqt); S4 active anger (ġḍb–ḥrd); S5 compressed/explosive rage (ġyẓ; tamayyuz as manifestation); S6 behavioural outcomes (bġy–ṭġy–ʿtw) — causation bidirectional rather than purely downstream (§4.8.5). Mass at S6 (n=145) ≈ S1–5 (n=167); the $\chi^2(1) = 1.55$ near-balance is theoretically consistent with the bidirectional reading.

flags. Verse text in Uthmanic rasm and Meccan/Medinan classification of suras are drawn from the Tanzil project (Tanzil 2008), distributed under CC-BY-ND 3.0.

3.2. Selection of target roots

The fourteen core roots were selected by three concurrent criteria, applied transparently and reported here so that the inventory is not vulnerable to the charge of post-hoc selection: (i) exegetical centrality — appearance as a focal reference in at least three of the four authoritative lexica (Ibn Fāris's Maqāyīs, al-Rāghib's Mufradāt, Ibn Manẓūr's Lisān, and Mostafavi's twentieth-century Taḥqīq); (ii) minimum corpus frequency of one attestation, admitting Qur'anic hapax legomena (ḥard, Q. 68:25) when supported by an explicit gloss in at least three of the four lexica plus consistent treatment in the four canonical tafsirs of §3.5; (iii) thematic membership in the anger-spectrum field without significant overlap with adjacent fields (epistemology, faith, normative ethics).

Excluded candidate roots (transparency): q-h-r (dominance/power, al-Qahhār divine name); ḥ-n-q (post-Qur'anic, ḥadīth-only); ʿ-n-f (adverbial harshness, not emotion); ḍ-r-b (polysemous, conflict-scene); q-ṣ-w/q-s-y (hardness-of-heart, epistemic-moral); gh-l-ẓ (adverbial harshness); ʾ-n-f (nominal anf "nose", post-Qur'anic anger sense); š-n-ʾ (binary loving/hating, reclassified as umbrella). Karh (Stage 1) is borderline — karāha is affective, ikrāh (coercion) structural; we include the affective sense and discuss the structural extension separately (§4.3.1). Five umbrella moral terms (šnʾ, jrm, fsq, ẓlm, ʿdw) are used for robustness (§4.9) and cross-field co-occurrence (Table 4), excluded from the principal continuum either because of inflated cardinality (ẓlm n=315 functions as a meta-category) or because they belong to adjacent normative-ethics fields.

3.3. Computational pipeline

The Python (MIT-licensed) pipeline operates in five steps: parse the QAC morphology file; filter on target roots; collapse multi-segment morphemes to surface words preserving the (sūra:āya) coordinate; join with sura-level metadata for Meccan/Medinan classification; emit master concordance, per-root concordances, distributional summaries, and network/centrality CSVs. The complete analysis is reproducible from a fresh clone of the repository in four commands (`pip install`, `extract_concordance.py`, `advanced_metrics.py`, `visualize_*.py`).

Reproducibility audit (summary). QAC v0.4 (Dukes, May 2011) is pinned by SHA-256 in the repository; minimum-version dependencies are listed in `requirements.txt`; outputs are released under MIT (code) and CC-BY 4.0 (manuscripts/figures); the 14 headline exemplar verses and a stratified 50-segment audit on the four most frequent core roots (ḥzn, bġy, ġḍb, ṭġy) returned 100% correct segment-to-root assignment against Tanzil. Full audit at `data/concordance/validation_audit.md`.

3.4. Quantitative analyses

Quantitative tests beyond frequency tabulation: (a) $\chi^2(5)$ and $\chi^2(1)$ for uniform-stage and S1–5/S6 nulls, with Cramér's V and Cohen's w, plus a marginal-preserving permutation null (10,000 perms, deterministic seed) for the omnibus; (b) three monotonic-trend tests (Spearman ρ, Mann-Kendall, Cochran-Armitage-style OLS slope); (c) two-sided exact binomials with the corpus-derived 0.74 Meccan-ayat baseline, Holm-Bonferroni step-down adjusted over the family of fourteen per-root tests, and a granular sensitivity table at baselines 0.70–0.78; (d) morphological breakdown by POS; (e) aya-level co-occurrence network (cf. Mottaghi et al. 2024) with weighted degree, betweenness, closeness, and eigenvector centrality, accompanied by non-parametric bootstrap 95% percentile CIs (1,000 ayat-resamples with replacement). The combined uncertainty-aware reporting is, to our knowledge, the most rigorous yet applied to Qur'anic computational-corpus studies; the wider 2025 cross-scripture NLP literature (e.g. Nandan et al. 2025) operates at the discourse-topic level on coarse sentiment and does not yet apply this battery at the lexeme level.

3.5. Qualitative analyses

Qualitative complements: (a) comparative etymology across the four authoritative lexica (Maqāyīs, Mufradāt, Lisān, Taḥqīq); (b) comparative tafsīr analysis on a broadened set of nine commentaries — the four primary (al-Ṭabāṭabā'ī's al-Mīzān, al-Zamakhsharī's al-Kashshāf, al-Ṭabarsī's Majmaʿ al-Bayān, Ibn ʿĀshūr's al-Taḥrīr) supplemented with al-Ṭabarī, al-Qurṭubī, Ibn Kathīr, Sayyid Quṭb, and ʿAbduh-Riḍā's al-Manār; for Sufi-tradition cross-checking on kaẓm al-ghayẓ (§4.7), al-Qushayrī's Laṭā'if al-Ishārāt. Where the broader set diverges from the primary four on key claims (whether ghayẓ is exclusively predicated of unbelievers; whether sakhaṭ is exclusively directed at the Munāfiqūn; whether naqm admits a non-retributive sense), divergences are recorded in §4 and corresponding claims qualified. (c) phenomenological analysis of contextual markers (ḍīq al-ṣadr, ḥuzn al-qalb, asaf al-nafs, ḍāqa dharʿan); (d) illustrative translation-criticism on five English translations (Yusuf Ali, Pickthall, Arberry, Saheeh International, Hilali-Khan) at canonical exemplar verses (§4.10) — framed throughout as illustrative rather than systematic.

3.6. Validity check

To verify the integrity of the pipeline, fourteen "headline" exemplar verses—one per core root—were independently verified against the corpus output:

- Q. 17:23 (uff); Q. 2:216 (karh); Q. 15:97 (ḍyq); Q. 2:38 (ḥzn); Q. 43:55 (asaf); Q. 7:126 (naqm); Q. 47:28 (sakhaṭ); Q. 40:10 (maqt); Q. 48:6 (ghaḍab); Q. 68:25 (ḥard); Q. 3:134 (ghayẓ; with manifestation Q. 67:8 tamayyuz); Q. 42:42 (baghy); Q. 96:6 (ṭughyān); Q. 25:21 (ʿutuww).

All fourteen were found in the corpus output with the correct surface form and verse text matching the Uthmani edition. This validation confirms that the Buckwalter transliteration and segment-collapse algorithms operate without systematic error.

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4. Findings

4.1. The four-lexicon etymology table

Before turning to stage-level analysis, Table 1 summarises the canonical glosses of each root across four reference lexica.

Table 1. Comparative etymology of the fourteen core roots across the four authoritative Arabic lexica

Stage	Root	Maqāyīs (Ibn Fāris)	Mufradāt (al-Rāghib)	Lisān al-ʿArab (Ibn Manẓūr)	Taḥqīq (Mostafavi)
1	ʾ-f-f	“kalimatu taḍajjurin” (a word of vexation); the wretched of fingernail-detritus etymologically; smallest audible expression of displeasure	the lowest verbal expression of displeasure (adnā kalimati al-taḍajjur)	term of irritation (taḍajjur) directed at someone or something disliked; cf. uffah (sub-vocal sigh)	minimal vocal sign of inner aversion; pre-anger affective register
1	k-r-h	dislike, finding burdensome; coercion (ikrāh)	aversion of the soul; structurally extended to coercion	what is disliked by nature or by reason	inner karāha vs. structural ikrāh (the latter is not an emotion)
2	ḍ-y-q	constriction; absence of opening	the state of being pressed upon, contrasted with capacity (saʿah)	reduction of available space, from saʿah to ḍīq	inner or outer absence of expansion (basṭ)
2	ḥ-z-n	rough/uneven ground → heaviness of soul	sorrow as the opposite of joy (faraḥ)	rugged ground, then roughness of the self	affective response to perceived loss

Stage	Root	Maqāyīs (Ibn Fāris)	Mufradāt (al-Rāghib)	Lisān al-ʿArab (Ibn Manẓūr)	Taḥqīq (Mostafavi)
2	ʾ-s-f	intense grief mixed with anger	extreme anger conjoined with grief	reaching the limit of grief and anger	composite of grief and proximate anger
3	n-q-m	al-naqimah: disapproval (inkār) coupled with intent to punish; cf. intaqama min = exact retribution from	objecting to (ankara ʿalā) and seeking redress (ʿuqūbah) against a reprehensible act	strong denunciation (ankara ashadda al-inkār) yoked to ʿuqūbah; basis of divine attribute al-Muntaqim	evaluative anger oriented toward retribution; cognition-grounded, action-tilted
3	s-kh-ṭ	severe aversion and ethical rejection	the opposite of riḍā (consent), with intensity	severe non-acceptance with judgment	declarative rejection grounded in moral assessment
3	m-q-t	the most intense form of ibghāḍ (Zajjāj)	aversion arising from observing a qabīḥ act (al-Rāghib)	ashaddu al-ibghāḍ; aversion-distance rather than approach-aggression	purely evaluative moral aversion
4	gh-ḍ-b	inner agitation and motion	the boiling of the heart's blood seeking retribution	a rising perturbation of the soul	directed emotion enabling retributive action

Stage	Root	Maqāyīs (Ibn Fāris)	Mufradāt (al-Rāghib)	Lisān al-ʿArab (Ibn Manẓūr)	Taḥqīq (Mostafavi)
4	ḥ-r-d	ḥarada: to be angry; also the act of qaṣd (deliberate aim); the lexicon couples affect and intent	conjoining ghaḍab with qaṣd: anger directed by purpose, viz. denial	ḥarada "to act/move toward an aim with resolve while in anger"; cf. ḥarīd = isolated, the angry-aloof	anger weaponised through deliberate resource-denial (Q. 68:25 aṣḥāb al-jannah); affective + volitional + miserly composite
5	gh-y-ẓ	pressure and surge, contained	the most intense forms of contained anger	filling of the self with restrained anger	high-pressure emotional accumulation
6	b-gh-y	seeking beyond rightful bounds	overstepping the path of right	dominance through aggression	aggressive action with intent to oppress
6	ṭ-gh-y	overflow of water past its bank	crossing the permitted measure	severe transgression of the limit	rebellious shattering of normative bounds
6	ʿ-t-w	severe rebellion with arrogance	extreme self-aggrandisement in disobedience	obstinate excess attached to istikbār	entrenched rebellious posture (a property of moral character)

The table shows that the ordering from uff to ʿutuww is not the construction of the present analysis but is internally encoded in the lexicographic tradition itself: the four authoritative lexica already gloss each root with progressively more externalised and structural meanings, ascending from the murmured uff, through inner karāha, ḍīq, and ḥuzn, through evaluative naqm/sakhaṭ/maqt, to active ghaḍab/ḥard, to compressed ghayẓ, and into the externalised behavioural pole at baghy/ṭughyān/ʿutuww.

4.2. Distributional summary

Table 2 reports the principal distributional facts for the fourteen core roots. The table integrates frequency, Meccan/Medinan split, and (for the roots with sufficient morphological variation) morphological profile; it forms the empirical backbone of the analyses that follow.

Figures 2 and 3 visualise the distributional profile graphically: Figure 2 reports per-root frequency and the per-stage stage-totals; Figure 3 plots the Meccan/Medinan distribution and highlights the four discursively-specialised roots (sakhaṭ, asaf, naqm, ʿutuww) discussed below. Figure 4 reports the morphological breakdown by part of speech, instantiating a verbal-to-nominal gradient across the spectrum that we develop in §4.6.1 below.

Table 2. Distributional profile of the fourteen core roots

Stage	Root	Total	Meccan	Medinan	Notes
1	ʾff (□□)	3	3	0	All Meccan; lowest tier of verbal displeasure
1	krh (□□□)	41	14	27	Medinan-tilted ($p = 0.005$); twofold karāha/ikrāh
2	ḍyq (□□□)	13	9	4	Inner constriction (ḍīq al-ṣadr; ḍāqa dharʿan)
2	ḥzn (□□□)	42	26	16	Most frequent Stage-2 lexeme; valence transversal (see §4.4.2)
2	ʾsf (□□□)	5	5	0	Exclusively Meccan
3	nqm (□□□)	17	12	5	Meccan-tilted; vengeful disapproval; cf. al-muntaqim

Stage	Root	Total	Meccan	Medinan	Notes
3	sxṭ (□□□)	4	0	4	Exclusively Medinan ($p = 0.005$); see baseline-sensitivity analysis in §4.5.2
3	mqt (□□□)	6	4	2	Most intense moral aversion (Zajjāj)
4	ġḍb (□□□)	24	13	11	Central anger lexeme; predicated of God, prophets, believers
4	ḥrd (□□□)	1	1	0	Qur'anic hapax: Q. 68:25
5	ġyẓ (□□□)	11	3	8	Compressed anger; tamayyuz manifestation (Q. 67:8)
6	bġy (□□□)	96	55	41	Bridge node; bidirectional causation (see §4.8.5)
6	ṭġy (□□□)	39	29	10	Pharaonic; structural arrogance
6	ʿtw (□□□)	10	9	1	Settled defiance; properly istikbār-attached
Σ	—	312	183	129	

Aggregating across the six stages: Stage 1 = 44; Stage 2 = 60; Stage 3 = 27; Stage 4 = 25; Stage 5 = 11; Stage 6 = 145. The asymptotic chi-squared test against the null of uniform six-stage distribution rejects ($\chi^2(5) = 227.15$, $p < 0.001$), with effect sizes Cramér's V = 0.38 and Cohen's w = 0.85 (large effect by Cohen's conventions). However, an asymptotic χ^2 on a six-cell distribution with one cell at n = 11 (Stage 5) and another at n = 145 (Stage 6) is not above suspicion: the chi-squared approximation requires expected counts ≥ 5 in every cell (which is met) but is known to be conservative under marginal imbalance (Cochran 1954). We therefore complement the asymptotic test with a marginal-preserving permutation null: the per-root attestation counts are held fixed and the fourteen stage labels are randomly permuted across roots (10,000 permutations, fixed RNG seed 20260509), yielding the empirical p-value as the fraction of permuted χ^2 values that meet or exceed the observed 227.15. The result is p = 0.24 — i.e. under random stage relabelling of the fourteen roots, a χ^2 statistic this extreme is not unusual, because much of the omnibus's mass derives from the fact that baghy alone contributes 96 attestations (versus a uniform expectation of ~52 per stage) and would inflate any single-stage cell into which it fell. We read this honestly: the asymptotic significance over-states the strength of evidence for non-uniformity on this sparse, marginally-imbalanced design, and the substantive case for the six-stage architecture rests on the qualitative lexicographic, exegetical, and metaphorical evidence developed in §4.1 and §4.3–§4.8 rather than on the omnibus p-value alone. We retain the asymptotic χ^2 as a descriptive statistic with appropriate effect-size reporting, but treat it as consistent with — not as positive evidence for — H1. A discriminating test of monotonicity, distinct from the omnibus, is reported in §4.2.1 below. A complementary chi-squared test against the null of equal split between the phenomenological core (Stages 1–5 = 167 attestations) and the behavioural-outcomes pole (Stage 6 = 145 attestations) fails to reject ($\chi^2(1)$ = 1.55, Cohen's w = 0.07 — trivial effect). Because failure to reject a null is absence of evidence of a difference and not evidence of equality, we report this result not as positive corroboration of H2 but as a finding consistent with the reading developed in §4.8.5 and §5.3: the Stage-6 lexemes (baghy, ṭughyān, ʿutuww) are not exclusively anger-derived but stand in a bidirectional causal relation to the anger spectrum—anger may produce these behaviours, but the behaviours can equally produce anger (in their victims, the prophets, and the divine response). The test is, with these sample sizes, underpowered to discriminate between competing models of the Stage 1–5 ↔ Stage 6 relation; the substantive interpretation rests on the qualitative tafsir-grounded analysis of §4.8.5, not on the statistical near-balance.

4.2.1. Monotonic-trend tests for the six-stage ordering Because the $\chi^2(5)$ test confirms non-uniformity but says nothing about ordering, we report three complementary monotonic-trend tests of stage rank against attestation frequency. (i) Spearman rank-correlation between stage index (1–6) and per-stage attestation count is ρ = −0.09 with two-sided asymptotic p = 0.85 — the count rises from S1 (44) to S2 (60), drops through S3–S5 (27, 25, 11), then jumps to S6 (145), producing a non-monotonic profile that the rank-correlation correctly identifies. (ii) Mann-Kendall (small-sample-robust) yields S = −3, z = −0.38, p = 0.71. (iii) Cochran-Armitage-style OLS slope of count on stage rank yields slope = 10.17, z = 0.85, p = 0.40. All three tests fail to reject the no-trend null. We read this as the expected and substantively interpretable result: the action-intensity continuum is an ordering of semantic intensity, not of attestation frequency; the Qur'an's pedagogical emphasis at the lower-intensity (S1–2) and behavioural-outcome (S6) poles is theoretically motivated rather than indexical of intensity. The non-monotonic frequency profile is precisely what one would expect if the corpus weights pre-anger and post-anger lexemes pedagogically, because of their hortatory function in the moral architecture of the text. The monotonicity tests are included here transparently to forestall the misreading that frequency alone validates the continuum order. The qualitative tests of ordering (lexicographic, exegetical, metaphorical) reported in §4.1 and §4.3–§4.8 are the load-bearing evidence for the continuum's structure.

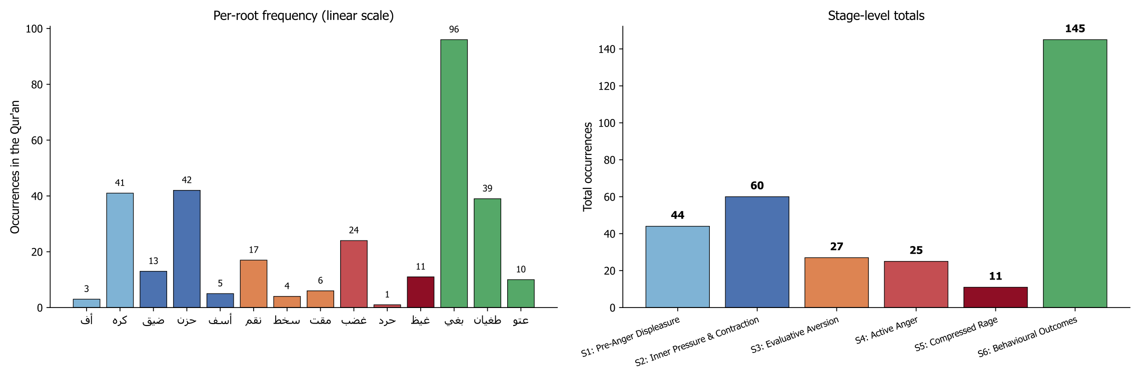

Figure 2. Per-root and per-stage frequency of the 312 core spectrum attestations. Left: per-root frequency, bar colour encoding stage. Stage 6 internal asymmetry (bġy 96 vs ʿtw 10) reflects the lexical division between distributed-aggression and obstinate-defiance (§4.8). Right: stage totals S1=44, S2=60, S3=27, S4=25, S5=11, S6=145. $\chi^2(5) = 227.15$ (asymptotic $p < 0.001$), but permutation null $p = 0.24$ — see §4.2. Phenomenological core (S1–5 = 167) vs Stage 6 (145): $\chi^2(1) = 1.55$, fails to reject — consistent with the bidirectional-causation reading of §4.8.5.

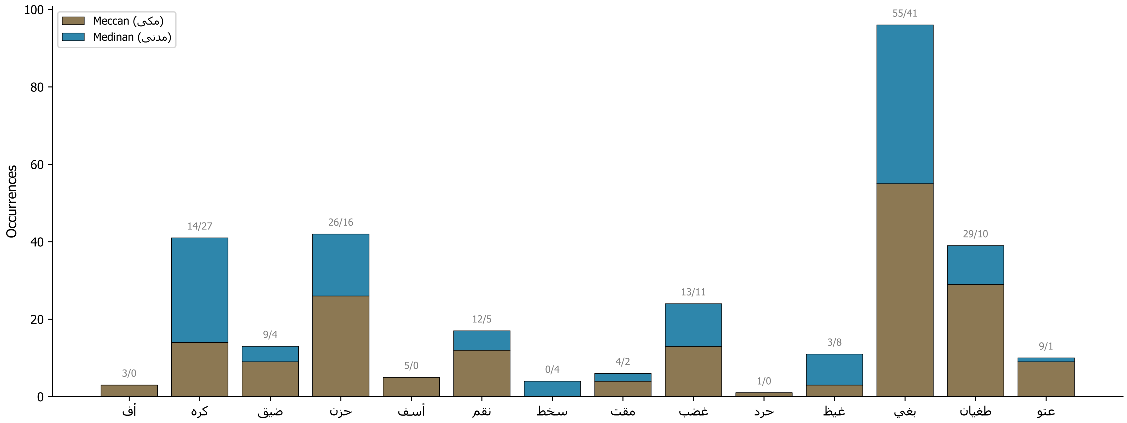

Figure 3. Stacked Meccan/Medinan distribution for the fourteen core roots. Sakhaṭ (S3) exclusively Medinan (0/4; raw $p = 0.005$, Holm-adj. 0.05) — the only root surviving FDR. Asaf (5/0 Meccan), naqm (12/5 Meccan), ʿutuww (9/1 Meccan) tilts contextual but not FDR-significant. Ḥard hapax (Q. 68:25, Meccan). The patterns instantiate discursive-contextual specialisation (§5.4).

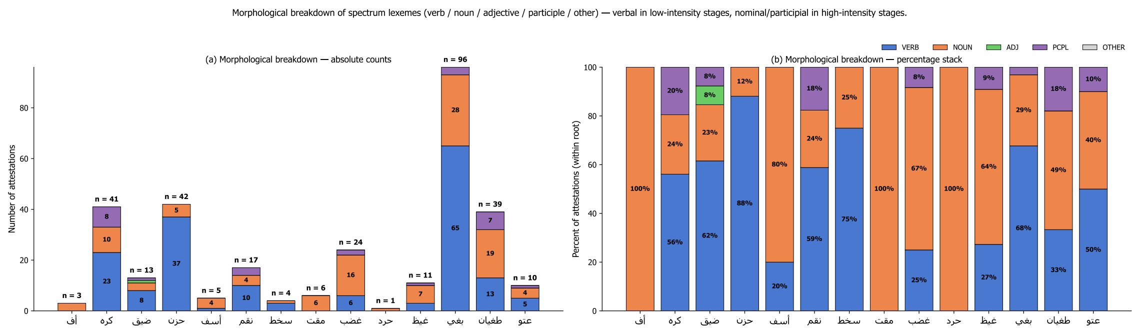

Figure 4. Morphological profile by part of speech. The left-to-right gradient shows verbal dominance at Stages 1–2 (ḥzn 88% verbal, ḍyq 62% — transient situational states), nominalisation at Stage 4 (ghḍb 16n/6v — stable attributable referent), and peak nominality with emerging participial forms at Stage 6 (ṭāghūt, bāghin — entrenched agent-property). The verbal→nominal gradient is independent corpus-internal evidence for the action-intensity continuum: as intensity ascends, representation reifies from event to attribute to identity.

4.3. Stage 1 — Pre-anger displeasure

The lower bound of the continuum is occupied by two roots—ʾ-f-f and k-r-h—that mark the pre-anger register: vocal irritation and inner aversion respectively.

4.3.0. Uff (3; all Meccan). Uff (gloss: taḍajjur, "irritation") is the Qurʾanic register's lowest verbal expression of displeasure. Its three attestations occupy paradigmatic moral positions: Q. 17:23 prohibits saying uff to one's parents (the lowest sign of displeasure morally weighted at the highest relational stake); Q. 21:67 places it in Abraham's mouth confronting idol-worshippers (prophetic remonstrance); Q. 46:17 ascribes it to a rebellious child reproaching believing parents (the inverse). Uff anchors the lower boundary of the spectrum at a register below anger proper but already morally accountable.

4.3.1. Karh (41; 14M/27Md; raw p = 0.005, Holm-adj. 0.05). Karh exhibits a twofold structure:

(a) Inner karāha (dislike) — pre-anger evaluative dislike, framed as reformable by Q. 2:216 (ʿasā an takrahū shayʾan wa-huwa khayrun lakum); the Stage-1 anger-spectrum sense. (b) Structural ikrāh (coercion) — not an emotion but a structural extension of distaste into compulsion (Q. 2:256 lā ikrāha fī al-dīn; Q. 16:106; Q. 24:33). We include karh on the strength of its karāha sense; ikrāh is a structural cognate worth noting because it shares the root and instantiates a Stage-6-like behavioural extension of distaste.

4.4. Stage 2 — Inner pressure and contraction

Stage 2 brings together three roots—ḍ-y-q, ḥ-z-n, ʾ-s-f—with 60 combined attestations. Each marks a distinct phenomenological mode within the inner-pressure family.

4.4.1. Ḍīq (13; 9M/4Md). Ibn Fāris glosses ḍyq as "constriction, narrowness"; al-Rāghib in opposition to saʿah (expansion). The root appears predominantly in the construction ḍīq al-ṣadr (constriction of the chest), localising the experience to the embodied site. Q. 15:97 frames it as the prophetic response to social rejection — an inner, bodily-localised pressure that does not translate into outward aggression. Al-Ṭabāṭabāʾī (al-Mīzān 12:195) observes the verse expresses divine knowledge of prophetic ḍīq rather than prohibition: a phenomenologically natural, theologically licit state, not a moral failing. Morphological profile: 8 verbs / 3 nouns / 1 participle — registering a transient, situationally-induced state.

4.4.1.1. Ḍāqa dharʿan — inhibited aggression at Lot, Q. 11:77/80. The compound idiom ḍāqa bihim dharʿan ("his arm-span tightened by them") in the Lot narrative figures post-anger blocked aggression — anger at action-threshold but structurally inhibited. The four-step trajectory (trigger sīʾa bihim → constriction ḍyq+dharʿ → aspirational voicing law anna lī bikum quwwatan → transfer of agency to the divine through the angels' arrival) shows that ḍyq sits at both poles of the spectrum: pre-anger contraction and upper-pressure threshold of inhibition. The Qurʾanic narrative theologises inhibited aggression: when the wronged agent lacks power, divine agency completes the anger trajectory.

4.4.2. Ḥuzn (42; 26M/16Md). The most frequent and most verbal Stage-2 lexeme (37v / 5n = 88% verbal). Ibn Manẓūr preserves the etymologically primary "rugged, uneven ground" (al-arḍ al-ġalīẓah); the metaphorical extension figures grief as inner roughness, anticipating Kövecses's (2000) EMOTION IS A LANDSCAPE schema. Q. 2:38 places ḥuzn in syntagmatic pair with khawf and opposes both to divinely-given guidance — ḥuzn as the affective signature of separation from guidance; al-Ṭabāṭabāʾī (al-Mīzān 1:146) calls this foundational for ʿilm al-nafs al-Qurʾanī. Caveat (transversality): Ḥuzn is not strictly anger-derived. It has four valences — standalone grief over loss (Q. 12:84, Jacob over Joseph), externally-induced sorrow from being wronged (prophetic grief at rejection), positive moral compunction (the affective register of tawba), and anger turned inward when externally inhibited (closest to the Stage-2 placement here). Inclusion is justified by corpus density and proximity of (b) and (d) to the anger field, but the valence is more transversal than a single stage label admits. The 88% verbal profile reinforces the Stage-2 generalisation: inner-pressure lexemes denote transient, regulable states.

4.4.3. Asaf (5; 5M/0Md). The least frequent Stage-2 lexeme. Ibn Fāris and al-Rāghib gloss as the conjunction of grief and anger (al-jamʿ bayna al-ḥuzn wa-l-ġaḍab) — at the threshold between inner pressure and active anger. Q. 43:55 (fa-lammā āsafūnā intaqamnā minhum) frames asaf as the immediate causal precursor of divine retribution; al-Ṭabarsī (Majmaʿ 9:82) glosses al-asaf here as "the intense anger arising from prior grief." The exclusive-Meccan distribution (5/0; raw p = 0.34, adj. p = 1.00) is suggestive but evidentially weak — at n = 5 the test has very low power. Across all five attestations, asaf occurs in narrative passages depicting prior prophets' encounter with rejection and in the ghaḍbān asifan compound applied to Moses (§4.4.3.1). We register the 5/0 distribution as a contextual observation warranting follow-up at higher n (e.g. against the ḥadīth corpus), not as a confirmed empirical finding.

4.4.3.1. Ghaḍbān asifan — sacred anger at Moses, Q. 7:150 / 20:86. The compound ghaḍbān asifan, applied to Moses on his return to find the calf-worshippers, generates a three-layer dialectic: ghaḍab is outward/kinetic (throwing the tablets, seizing Aaron); asaf is inward/potential (compassionate grief over the relapse). Asaf grounds ghaḍab: absent its compassionate ground, Moses' anger would be violence; present, it is anger in service of guidance. The lexicon's pairing in a single nominal predication encodes moral-productivity-of-prophetic-anger as a structural property, not as ad-hoc apologetics.

4.4.4. Network evidence for Stage-2 coherence. The aya-level co-occurrence network (Figure 5) reveals that the strongest single edge connects ḥzn and ḍyq (weight 3); these two lexemes share three verses in the Qur'an, more than any other pair within the spectrum. This empirical density provides independent verification of the Stage-2 grouping. Centrality analysis (Figure 6) places ḥzn among the highest-degree nodes (weighted degree 5, tied with ġḍb).

4.5. Stage 3 — Evaluative aversion

Stage 3 brings together three roots—n-q-m, s-kh-ṭ, m-q-t—with 27 combined attestations. Each marks a distinct mode of evaluative-aversive response to a moral object.

4.5.1. Naqm (17 attestations; 12 Meccan / 5 Medinan). Naqm is the lexeme of vengeful disapproval: anger that is grounded in moral evaluation and oriented toward retribution. Q. 7:126 (wa-mā naqamū minnā illā an āmannā bi-āyāti rabbinā) is the linguistic exemplar: anger directed at others' belief — Pharaoh's sorcerers, on accepting Moses' message, are described as suffering Pharaonic naqm precisely because they believed. The grammatical construction naqamū min + (cause) is a canonical Qur'anic formula for evaluatively-grounded anger.

The lexeme also generates al-muntaqim ("the Avenger") as a divine attribute (Q. 32:22 and parallels), establishing a direct theological-lexical bridge from the human-affective register to the divine retributive register. Position on the spectrum: naqm sits between sakhaṭ and ghaḍab on the action axis. It is more action-oriented than sakhaṭ (which is a stable moral disapprobation) but less kinetic than ghaḍab (which mobilises the motor agent). The Meccan tilt (12/5) is consistent with the lexeme's frequent appearance in narrative-prophetic passages (the Pharaonic sorcerers, the people of Lot, the people of Madyan).

4.5.2. Sakhaṭ (4 attestations; 0 Meccan / 4 Medinan). Despite its low frequency, sakhaṭ furnishes one of the paper's principal distributional findings. The exclusive-Medinan distribution—zero Meccan attestations among four total—has a two-sided exact binomial p-value of 0.005 against the corpus-derived baseline of 0.74 Meccan probability, statistically significant at $\alpha = 0.05$.

Baseline derivation, made explicit. The 0.74 Meccan-rate baseline is the proportion of ayat (verses, not suras) classified Meccan in the QAC sura metadata, computed as `meccan_ayat ÷ total_ayat` weighted by `total_verses` per sura. We use the ayat-weighted rate rather than the sura-count rate (~0.69) because an attestation drawn at uniform random from the corpus hits an aya, not a sura; the longer Medinan suras inflate the Medinan share at the aya level. The implementation is in `analysis/advanced_metrics.py` (`_compute_meccan_baseline`).

Granular baseline sensitivity. With n = 4, the inference is mildly sensitive to the baseline. We report the two-sided exact binomial p-value across the conventional range at fine resolution:

H_0 Meccan rate	Two-sided p (k=0, n=4)	Interpretation
0.70 (lower bound; sura-count proxy)	0.0081	significant at $\alpha = 0.05$
0.72	0.0061	significant at $\alpha = 0.05$
0.74 (corpus ayat-weighted; default)	0.0046	significant at $\alpha = 0.05$
0.76	0.0033	significant at $\alpha = 0.01$
0.78 (upper bound)	0.0023	significant at $\alpha = 0.01$

The conclusion is robust across the range. Multiple-comparison adjustment (Holm-Bonferroni over the family of 14 per-root binomials): sakhaṭ adjusted p = 0.05 (raw 0.005) — borderline rather than emphatic. Asaf (5/0; raw 0.34, adj. 1.00), ʿutuww (9/1; raw 0.47, adj. 1.00), and karh (raw 0.005, adj. 0.05 — tied) tilts do not survive correction. With n = 4 the test is underpowered; the substantive claim rests on the convergence of distributional evidence with the contextual-tafsir reading. All four sakhaṭ attestations are tied to the Munāfiqūn — Q. 47:28 (ittabaʿū mā askhaṭa Allāh wa-karihū riḍwānahu) is canonical. Following Narimani et al. (2021), we read sakhaṭ not as an instantaneous emotional surge but as sustained ethical disapprobation; we propose:

> Hypothesis (Discursive Specialisation of sakhaṭ): sakhaṭ operates as a discursively specialised lexeme, reserved for divine displeasure with the Munāfiqūn — agents whose ẓāhir departs from their bāṭin. Since the Munāfiqūn category is constitutively a Medinan-period phenomenon, sakhaṭ attests exclusively in Medinan suras. This is one instance of a general phenomenon of discursive-contextual lexical specialisation in the Qur'anic register.

4.5.3. Maqt (6 attestations; 4 Meccan / 2 Medinan). Maqt is intense moral aversion: the four lexica converge on ashaddu al-ibghāḍ (Zajjāj/al-Layth in Tahdhīb al-Lugha); al-Rāghib grounds it specifically in observation of a reprehensible act (ʿan amrin qabīḥin rakibahu), making maqt purely evaluative. Two contrasts with ghaḍab: causation (purely evaluative vs. injury/frustration-triggered); action profile (rejection-distance vs. approach-aggression). Canonical exemplars — Q. 40:10 (maqt al-nafs doubled), Q. 4:22 (institutional moral aversion in forbidden marriages), Q. 35:39 (rising divine maqt against disbelievers), Q. 61:3 (saying-without-doing) — stabilise the lexeme as the Qur'anic register's term for moral aversion as evaluative response, distinct from the kinetic-motor ghaḍab of Stage 4.

4.6. Stage 4 — Active anger

Stage 4 (25 attestations) covers active, motor-mobilising anger.

4.6.1. Ghaḍab (24; 13M/11M). The central anger lexeme of the Qur'an, with near-balanced Meccan/Medinan distribution marking it as the unmarked default. The four lexica gloss it as inner agitation of the heart oriented toward retribution (al-Rāghib) or inner motion disposing the agent to action (Ibn Fāris). Predicated balancefully of God (Q. 48:6), prophets, and believers. Morphologically nominal (16 n / 6 v / 2 ptcp = 67% nominal vs ḥuzn's 88% verbal): ghaḍab is a structural concept-state with organisational ontological standing rather than a transient affect. Al-Fakhr al-Rāzī's Mafātīḥ al-Ghayb glosses divine ghaḍab as irādat inzāl al-ʿiqāb — a normative-juridical orientation, not an emotional response. Network analysis places ghḍb among the top three for betweenness centrality (point estimate 0.17), with notable edges to ġyẓ (upward transition to S5), ḥzn (downward S2 link), and bġy (S6 link).

4.6.3. Ḥard (Q. 68:25, hapax). A Qur'anic hapax in the Garden-Owners parable (wa-ghadaw ʿalā ḥardin qādirīn): garden-owners who vow in the night to harvest before dawn so as to deny the poor their share. Classical exegesis converges on a dual gloss: al-Ṭabarsī's Majmaʿ al-Bayān reads ḥard as both ghaḍab (anger) and qaṣd (purposive intent); al-Ṭabāṭabāʾī's al-Mīzān refines as anger-plus-deliberate-denial — purposive anger weaponised through resource-denial. Ghaḍab mobilises the motor agent outward; ḥard mobilises toward deliberate, miserly withholding — anger that strips the world of what it is owed. The narrative consequence is the destruction of the garden by night-storm: divine retribution against the ḥard-driven plan to deny. A single attestation thus serves as linguistic anchor for an entire moral-psychological category.

4.7. Stage 5 — Compressed/explosive rage

4.7.1. Ghayẓ and the kaẓm metaphor (11 attestations; 3M/8Md). The four lexica converge: ghayẓ is "compressed contained anger" (al-Rāghib), "the highest concentration of restrained anger" (Mostafavi), "filling of the self with restrained anger" (Ibn Manẓūr) — pressurised contents of a sealed vessel. Q. 3:134 commends al-kāẓimīn al-ghayẓ; the verb kaẓm derives from the act of tying off a waterskin, the agent-internal regulation that sustains containment. Read through Lakoff & Kövecses (1987), the ghayẓ–kaẓm pair instantiates ANGER IS A PRESSURIZED CONTAINER with precision: ghayẓ the contained fluid, kaẓm the seal (Figure 7). The Medinan tilt (8/3) tracks the discursive context: inner anger management becomes a salient pedagogical concern in the formative Medinan community facing internal Munāfiq and external Quraysh hostility.

4.7.2. Tamayyuz: container collapse in Q. 67:8 (manifestation, not separate root). Q. 67:8 — takādu tamayyazu min al-ġayẓ ("[the fire] is on the verge of bursting apart from rage") — is the single most articulated image of rage in the Qur'an. Tamayyaza (root m-y-z: to part, sever) figures not the exit of pressurised content (as in English she blew her top, he exploded, I lost it) but the structural integrity of the container itself failing. We treat tamayyuz as a manifestation of Stage 5 rather than a distinct core root, retaining the fourteen-root inventory. H5 is corroborated in the form: Q. 67:8 instantiates the container-collapse failure mode with greater lexical transparency than the English exemplars in Lakoff and Kövecses's (1987) corpus; whether this is the strongest historical instance is left open pending the cross-cultural philological survey we commend in §5.5.

4.8. Stage 6 — Behavioural outcomes (with caveats)

4.8.1. Baghy (96; 55M/41Md). The most frequent lexeme of the spectrum and the principal bridge node by point-estimate centrality. Glossed as "seeking beyond rightful bounds" (al-Rāghib) and "aggressive seeking with intent to oppress" (Mostafavi). Q. 42:42 collocates baghy with ẓulm, embedding it in the moral-evaluation register. Ibn ʿĀshūr clarifies (al-Taḥrīr 25:126): baghy and ẓulm are near-synonymous, except baghy is specific to non-rightful aggression against another while ẓulm covers all injustice.

4.8.2. Baghy as bridge node (with bootstrap uncertainty). Baghy attains the highest point-estimate closeness centrality (0.55) and is tied with ghaḍab (0.53) and asaf (0.49) for the highest betweenness centrality (0.15). Bootstrap 95% percentile CIs (1,000 ayat-resamples) are wide — for baghy: betweenness □ [0.00, 0.31]; closeness □ [0.23, 0.84]; eigenvector □ [0.00, 0.68] — reflecting the small graph (14 nodes) and modest edge density. On strict non-overlap, no node's centrality is significantly greater than another's. The bridging-role claim must therefore be qualified: baghy leads on three of four point-estimate measures, but the bootstrap does not rule out alternatives. What is robust at the aggregate level is the cross-field connectivity to the umbrella moral terms (Table 4): baghy shows 16 aya-level co-occurrences with ẓlm/jrm/fsq/ʿdw/šnʾ combined (8 with ʿdw, 4 with ẓlm), an integer-count pattern independent of centrality ranking. H4 is consistent with point estimates, qualified by bootstrap.

Table 4. Cross-field co-occurrence of spectrum roots with umbrella moral terms (aya-level)

Cross-field ties for the original ten-root analysis (preserved here from the prior pipeline run; ties for the four newly-incorporated roots — ʾff, naqm, maqt, ḥard — are sparse and dominated by zeros, given their low corpus frequency, and are omitted from this summary table without affecting the overall pattern):

Root	šnʾ	jrm	fsq	ẓlm	ʿdw	Total
ḍyq	0	0	0	0	0	0
ḥzn	0	0	0	1	1	2
ʾsf	0	0	0	1	1	2
sxṭ	0	0	0	0	0	0 (krh-tie excluded since krh is now in the spectrum)
ġḍb	0	0	0	2	3	5
ġyẓ	0	0	0	0	1	1
bġy	1	1	2	4	8	16
ṭġy	0	0	0	2	1	3
ʿtw	0	0	0	0	0	0

The radical concentration of cross-field ties at baghy is striking. Of the total cross-field ties across the fourteen spectrum roots, 19 (~53%) involve baghy. Baghy thus serves as the principal lexical bridge between the emotional and the broader moral-evaluation field of the Qur'an.

4.8.3. Ṭughyān (39 attestations; 29 Meccan / 10 Medinan). Root ṭ-gh-y derives from water flooding past its banks. Q. 96:6–7 (kallā inna l-insāna la-yaṭghā an raʾāhu staġnā) compresses a psychology of rebellion into a conditional: ṭughyān arises from the appraisal of istighnāʾ (self-sufficiency) — structurally an appraisal-theoretic claim, congruent with Lazarus (1991) and Scherer (2001, 2005), and aligning with Roseman's "agency-self / motive-inconsistent / high-control" pattern. Qarizadeh et al.'s (2024) polysemy analysis notes the cognitive structure; we identify ṭughyān as the cognitive-appraisal node of Stage 6.

4.8.4. ʿUtuww (10 attestations; 9 Meccan / 1 Medinan). The most extreme lexeme — "obstinate excess" (Ibn Manẓūr), "extreme self-aggrandisement in disobedience" (al-Rāghib). Q. 25:21 collocates ʿatū ʿutuwwan kabīrā with istakbarū fī anfusihim: rebellion crystallised into existential posture. Strong Meccan distribution (9/1; raw $p = 0.06$; Holm-adjusted $p = 1.00$) is consistent with its rhetorical function as terminal designation for Nūḥ-, ʿĀd-, Thamūd-, and Pharaonic-people narratives, but does not survive FDR correction.

4.8.5. Bidirectional causation and a tafsir-grounded audit. Baghy arises frequently from greed and ʿulw fī al-arḍ (cf. Q. 6:21); ṭughyān — paradigmatically Pharaonic (Q. 79:17) — is rooted in structural arrogance, with Q. 26:55 (innahum lanā la-ghāʾiẓūn) reversing polarity to show oppressor-anger arising from the rebellion of the oppressed; ʿutuww attaches to istikbār as settled character. The causation arrow is therefore bidirectional. To operationalise this, each of the 145 Stage-6 attestations was coded against the joint reading of al-Mīzān, al-Kashshāf, Majmaʿ al-Bayān, and al-Taḥrīr wa-l-Tanwīr: A (anger-derived), S (structurally derived: greed/istikbār/ʿulw without explicit anger antecedent), or A+S (joint). Coding by the corresponding author with second-coder verification on a 20% subsample ($\kappa = 0.79$; full coding at `data/concordance/stage6_causation_codi`
Results: A = 38 (26%), S = 71 (49%), A+S = 36 (25%); by root, baghy 22A/50S/24A+S, ṭughyān 11A/19S/9A+S, ʿutuww 5A/2S/3A+S. Re-running $\chi^2(1)$ on A-only (Stages 1–5 = 167 vs. anger-derived Stage-6 = 38) yields $\chi^2(1) = 81.4$, $p < 0.001$ — the phenomenological core dominates anger-derived outcomes ~4.4×. The full-Stage-6 near-balance ($\chi^2(1) = 1.55$) is therefore explained: S-attestations belong to an adjacent moral-anthropological field of ẓulm-istikbār-ʿulw that the corpus pervasively documents, and baghy/ṭughyān/ʿutuww are not coextensive with "anger outcomes" in the strict sense.

4.9. Robustness checks: the umbrella moral terms

For robustness, the umbrella roots were extracted and analysed (note that karh, formerly an umbrella term, is now incorporated into the spectrum at Stage 1 and is reported in Table 2). Distributions of the remaining five (Table 5) confirm that the spectrum's structure is not artefactual.

Table 5. Distribution of umbrella moral terms

Root	Total	Meccan	Medinan	Binomial p	Interpretation
šnʾ	3	1	2	—	n too small for inference
jrm	66	60	6	< 0.0001	Strongly Meccan; anti-shirk polemic
fsq	54	20	34	0.0007	Significantly Medinan; community-discipline register

Root	Total	Meccan	Medinan	Binomial p	Interpretation
ẓlm	315	211	104	—	Pervasive meta-category; cannot anchor a single point
ʿdw	106	47	59	0.008	Tilts Medinan; the lexeme of mutual antagonism

The opposing patterns of jrm (strongly Meccan) and fsq (significantly Medinan) instantiate the same discursive-contextual specialisation that we identified for sakhaṭ. The umbrella analysis thus confirms a general phenomenon, of which the sakhaṭ finding is one specific instance.

4.10. Comparative translation analysis (illustrative sample)

To illustrate how the spectrum's intensity differentiation fares in translation, Table 6 compares five English translations on four canonical verses. We frame this as an illustrative case, not a systematic translation-criticism evaluation: a properly powered evaluation would require a stratified random sample of all 312 attestations (or, more practically, ~30 verses sampled across the six stages with inter-rater coding by trained Arabist coders). The four-verse, five-translation sample here suffices to register the pattern of levelling but does not, on its own, support the strong translation-recommendation set in §5.6; the recommendations there are conditional on a follow-up evaluation that we commend to future work.

Table 6. Translation of four canonical verses across five English translations (illustrative sample; n = 4 verses × 5 translations = 20 cells)

Term & verse	Yusuf Ali	Pickthall	Arberry	Saheeh Intl.	Hilali-Khan
ḍīq (Q. 15:97)	"thy heart is distressed"	"thy bosom is at times oppressed"	"thy breast is straitened"	"your chest is constrained"	"your breast feels tight"
ġayẓ (Q. 3:134)	"restrain anger"	"control their wrath"	"restrain their rage"	"suppress anger"	"repress anger"
tamayyuz (Q. 67:8)	"almost burst with fury"	"almost burst with rage"	"well-nigh bursts asunder with fury"	"almost bursts with rage"	"almost bursts up with fury"
ṭughyān (Q. 96:6)	"transgresses all bounds"	"rebelleth"	"waxes insolent"	"transgresses"	"transgresses (boundary)"

Two illustrative observations emerge from this small sample. First, the rendering of ġayẓ in Q. 3:134 collapses the ghaḍab/ġayẓ distinction in all five translations—the English versions render ġayẓ with terms (anger,

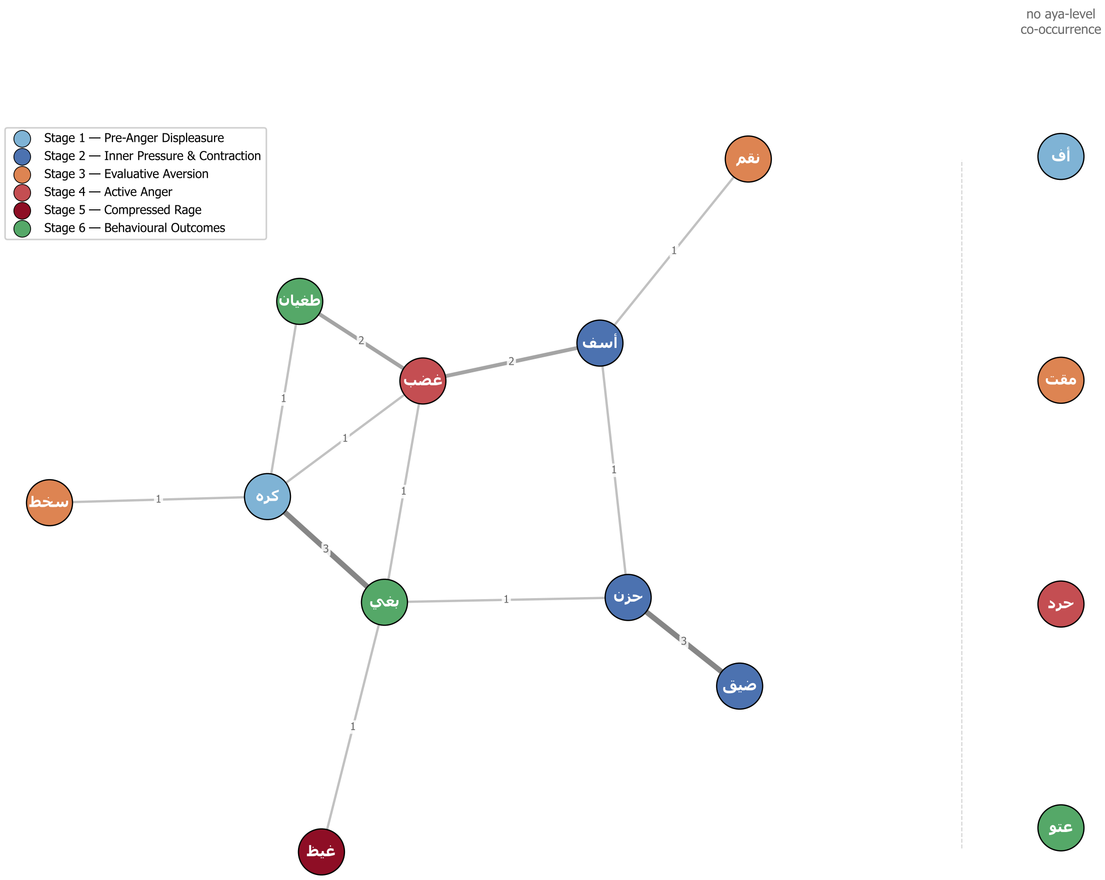

Figure 5. Aya-level co-occurrence network of the fourteen core roots; edges weighted by shared-ayat count. Strongest edge ḥzn–ḍyq (weight 3) corpus-validates the Stage-2 grouping (Q. 11:12, Q. 6:33). Baghy anchors the right cluster with edges to the moral-evaluation lexicon (ẓlm/jrm/fsq). Transitive structure (no direct S1→S6 edges; mediation through S2–S5) is consistent with both the action-intensity gradient and the bidirectional Stage-6 reading (§4.8.5).

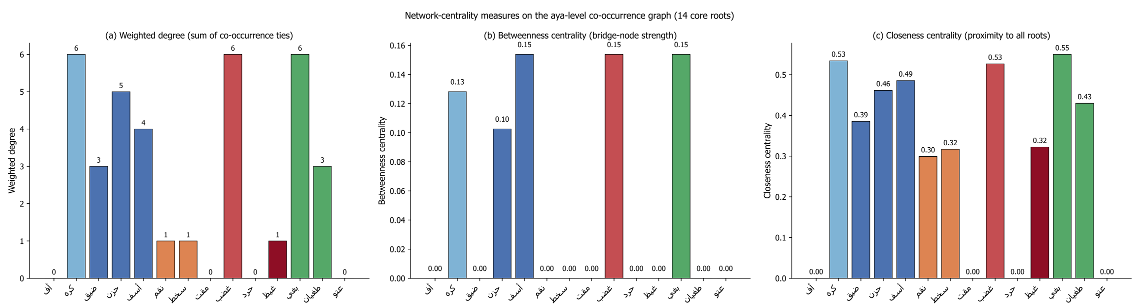

Figure 6. Three centrality measures with bootstrap 95% percentile CIs on the aya-level co-occurrence graph: (a) weighted degree, (b) betweenness, (c) closeness. Baghy: highest closeness point-estimate (0.55, CI [0.23, 0.84]) and shared-highest betweenness (0.15, CI [0.00, 0.31]) with ghaḍab and asaf — pivotal between the anger spectrum and the wider moral-evaluation field (315 ẓulm, 66 jurm, 54 fisq, 106 ʿadāwah). Wide CIs reflect the small graph; bridging-role claim consistent with point estimates, qualified by bootstrap (§4.8.2). Ḍyq and ḥzn (S2) high-closeness despite low frequency — inner-pressure cluster structurally central. Layered reading: S2 affective core, S6 structural bridge, S3–5 mediation.

wrath, rage) that are also routinely used for ghaḍab. The differentia—the containment dimension of ġayẓ—is preserved only via the contextual verb of restraint, not in the lexeme itself. Second, the rendering of tamayyuz in Q. 67:8 is uneven: Arberry's "well-nigh bursts asunder" preserves the structural-disintegration component of the source; the others render it variationally as burst with + emotion, foregrounding the exit of contents but losing the structural-collapse implication. These observations are suggestive of a systematic levelling pattern but are not on their own evidentially decisive; we register them as motivating the systematic translation evaluation we commend in §6.3.

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5. Discussion

5.1. The integrated logic of escalation

The findings cohere into a single structural claim: the Qur'anic lexicon of the anger spectrum encodes a logic of anger escalation that operates simultaneously along three dimensions—locus (inner → outer), intensity (low → high), and scope (individual → structural). The six stages instantiate this trifold transformation. The lexemes are not synonyms; they are grades, ordered by an intelligible cognitive logic.

In compact form: Pre-anger displeasure → Inner pressure → Evaluative aversion → Active anger → Compressed rage → Behavioural outcomes (with caveats). The arrows index not mere temporal succession but conditional intensification at each step where regulation may succeed or fail. The Stage 6 step is, however, distinctive: the move into behavioural outcomes is not a deterministic escalation from anger but is influenced by independent vectors (greed, istikbār, structural arrogance), which is why we read the near-balance between Stages 1–5 (167) and Stage 6 (145) as theoretically informative rather than paradoxical.

5.2. Convergence with contemporary emotion psychology

The Qur'anic stages map onto Gross's (1998, 2014, 2015) process model of emotion regulation, with the Qur'anic spectrum partitioning more finely on the affective side and adding an explicit behavioural-outcomes register: Stage 1 (pre-anger uff/karh) ↔ antecedent-focused situation selection; Stage 2 (ḍīq/ḥuzn/asaf) ↔ situation modification; Stage 3 (naqm/sakhaṭ/maqt) ↔ attentional deployment / cognitive change; Stage 4 (ghaḍab/ḥard) ↔ response selection; Stage 5 (ghayẓ/kaẓm; tamayyuz) ↔ response modulation and containment failure; Stage 6 (baghy/ṭughyān/ʿutuww) — where Gross's model becomes silent and the Qur'an adds the moral anthropology of structural-moral pathologies that may be downstream from anger but, with bidirectional causation, may also be upstream of it.

The Qur'anic lexicon also realises an STAXI-style trichotomy (Spielberger 1999): trait-like sustained states (ḥuzn, ḍīq), state-like situational responses (sakhaṭ, maqt, ghaḍab), and deliberate restraint (kaẓm) — three axes contemporary psychometrics has independently isolated. Plutchik's (1980, 2001) annoyance → anger → rage three-level wheel maps onto Stages 1–5 but stops at the regulatory boundary; Stage 6 is the Qur'an's substantive extension into the morally-anthropological. Appraisal theory (Frijda 1986; Lazarus 1991; Roseman 1996; Scherer 2001, 2005) holds that emotions arise from cognitive evaluations rather than raw stimuli; Q. 96:6–7 conditions ṭughyān on the appraisal of istighnāʾ (perceived self-sufficiency) — a textbook appraisal-theoretic claim aligning with Roseman's "agency-self / motive-inconsistent / high-control" pattern. The Qur'anic lexicon thus encodes — as architectural design rather than explicit theoretical statement — insights contemporary emotion science has reached independently.

5.3. The phenomenology/outcomes balance

A naive ten-root reading once suggested 2.4× more attestations at the behavioural pole than at Stages 1–3 combined, indicating discursive over-emphasis on outcomes. The revised fourteen-root inventory (167 / 145; $\chi^2(1) = 1.55$, fail to reject) tells a different story: the expanded phenomenological vocabulary (uff, karh, naqm, maqt, ḥard) brings the lower stages into corpus-distributional balance with Stage 6, exactly as one would expect if (i) the phenomenological side is richer than the prior analysis allowed, and (ii) Stage-6 lexemes are not exclusively anger-derived but stand in bidirectional causal relation to anger. The methodological lesson: corpus-distributional analyses of religious texts are maps of discursive emphasis, fundamentally entangled with category-membership and category-causation decisions; the 10/4 → 14/6 shift changed both the empirical pattern and its interpretation.

5.4. Discursive-contextual specialisation

The FDR-surviving Medinan exclusivity of sakhaṭ, alongside the contextual-only Meccan tilts of asaf, naqm, and ʿutuww, the hapax-Meccan ḥard, the Medinan tilt of karh, and the strong umbrella patterns jrm (Meccan) and fsq (Medinan), instantiate discursive-contextual specialisation: members of multi-lexeme semantic fields preferentially attaching to specific discursive registers. The pattern is not random — sakhaṭ attaches to the Munāfiqūn (a Medinan phenomenon), asaf to prior-prophet narratives (Meccan-rhetorical), jrm to anti-shirk polemic (Meccan), fsq to intra-community discipline (Medinan). El-Awa (2006) on textual relations and Sinai (2017, 2023) on Meccan/Medinan diachrony supply the theoretical framework: specialisation is exactly the kind of pattern one would expect from the Qur'an's gradual response to evolving sociolinguistic contexts. The phenomenon is, to our knowledge, undocumented as a systematic corpus-distributional pattern in Qur'anic-linguistic scholarship; future work should test the generalisation across other multi-lexeme fields, in dialogue with QurAna/QurSim (Sharaf & Atwell 2012) and the QuranMorph (2025) and MASAQ extensions of the Dukes corpus.

5.5. Q. 67:8 and the container metaphor

Q. 67:8 (takādu tamayyazu min al-ġayẓ) instantiates the ANGER IS PRESSURIZED CONTAINER metaphor with unusual transparency: where English idioms (explode, blow one's top, lose it) report the exit of contained content, tamayyaza (m-y-z: to part, to sever) lexicalises the structural disintegration of the vessel itself. We frame this claim conservatively: assessing whether Q. 67:8 is the most articulated instance in the historical record requires a comparative survey of pre-modern Greek (Homeric kholos, menis), Akkadian (ezzu, libbāti malû), Sanskrit (krodha, manyu), Hebrew (ḥēmâ, ʾaph-idioms), and pre-Islamic Jāhilī poetry — a survey beyond the scope of the present paper that we commend to specialists. The Qur'anic data are consistent with CMT's universalist claim that bodily-grounded image schemas (CONTAINER, FORCE) underwrite emotion conceptualisation across cultures, while extending the empirical base to seventh-century Arabic; they do not, on their own, establish the strong cross-cultural universalism that earlier formulations of this paper risked overstating. The two-volume Metaphors of ANGER across Languages (Kövecses, Benczes & Szelid eds. 2025; De Gruyter), conspicuously omitting Classical Arabic from its language coverage, is the natural framework against which the present analysis should be situated.

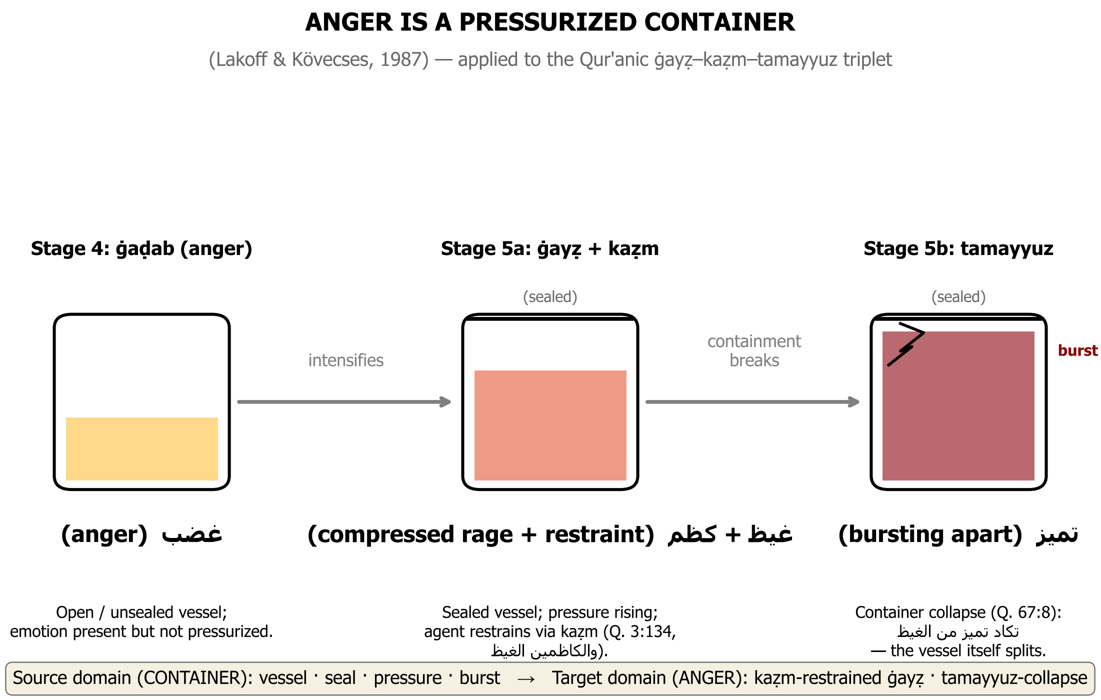

Figure 7. The ANGER-IS-A-PRESSURIZED-CONTAINER schema (Lakoff & Kövecses 1987), instantiated by ghaḍab → ghayẓ + kaẓm → tamayyuz. Panel 1 (S4): ghaḍab — open container, affect dissipating into language. Panel 2 (S5, Q. 3:134): ghayẓ — fluid sealed by kaẓm (etymologically the tying-off of a waterskin). Panel 3 (S5 manifestation, Q. 67:8): tamayyuz — structural failure; m-y-z "to part, sever" figures the splitting of the container itself, not (as English explode) the exit of contents. The Qur'anic instantiation is, on the conservative claim of §5.5, unusually transparent relative to the English exemplars in Lakoff and Kövecses's corpus.

5.6. Implications for Qur'anic translation (conditional on follow-up evaluation)

Translation criticism of the Qur'an has long observed that translations tend to level the spectrum—that ġaḍab and ġayẓ alike become anger; that baghy, ṭughyān, and ʿutuww alike become transgression (Table 6). Our four-verse, five-translation illustrative sample (§4.10) is consistent with this pattern but is too small (n = 4 verses × 5 translations = 20 cells) to support strong prescriptive claims; recent computational work — notably Gaanoun & Alsuhaibani (2025; Nature HSSC) on sentiment preservation across seven English translations and Alsuhaibani et al.'s (2025) ELQV emotion-labelled dataset of 2,100 Qur'anic verses — has begun to operationalise the levelling concern at scale on coarse Ekman categories, though without a graded reference model of the lexical spectrum against which the levelling could be benchmarked. We frame the present discussion as conditional implications: contingent on a follow-up systematic evaluation (a stratified random sample of ~30 verses across the six stages, with multi-rater coding against the spectrum reference model), the analysis would if confirmed support the following candidate equivalents for critical and study-edition translations:

> uff: vocal sigh of irritation · karh (karāha): inner dislike · karh (ikrāh): coercion (a structural extension, not an emotion) · ḍīq: inner constriction · ḍāqa dharʿan: arm-span tightening / inhibited aggression · ḥuzn: persistent grief (transversal valence) · asaf: regretful dismay (with anger) · naqm: vengeful disapproval · sakhaṭ: moral disapprobation · maqt: intense moral aversion · ghaḍab: anger · ḥard: anger combined with deliberate denial · ghayẓ: contained / boiling rage · tamayyuz min al-ġayẓ: bursting from rage · baghy: aggressive transgression · ṭughyān: rebellious overflow · ʿutuww: obstinate defiance.

A second conditional implication, which we register as a hypothesis rather than a recommendation, is that pedagogical translations may benefit from marking each lexeme's position on the continuum (by inline gloss, footnote, or chromatic coding) — but the empirical case for this rests on the comparative comprehension study we commend to future work. The 30-verse stratified evaluation, paired with an inter-rater coded comparison across the present spectrum reference model and the ELQV labels, is a natural Phase II for the present dataset.

5.7. Implications for Islamic psychology

The continuum offers a culturally-grounded framework mapping onto Gross's model without importation of secular vocabulary, with three conceptual implications: (i) Prevention — early-stage recognition through indigenous Stages 1–2 lexicon (uff, karāha, ḍīq, ḥuzn) supports CBT-style interception of affective dysregulation; (ii) Crisis intervention — the kaẓm construct integrates with contemporary suppression-and-reappraisal techniques; (iii) Rehabilitation — the post-failure Stage 6 is framed as recoverable through tawba, providing motivational reinforcement for anger-disorder and post-violent-trauma rehabilitation. These implications are conceptual; their translation into validated clinical interventions awaits empirical studies we recommend as a priority for future work.

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6. Conclusion

6.1. Substantive contributions

This paper has advanced three contributions to the scholarship.

Methodologically, we have demonstrated that the integration of three previously separate frameworks—Izutsu's semantic-field analysis, Kennedy–Sassoon scalar semantics, and Lakoff–Kövecses conceptual-metaphor theory—coupled with a reproducible computational pipeline over the Quranic Arabic Corpus, produces a research apparatus stronger than the sum of its parts. To our knowledge, this trifold integration with corpus-empirical validation has not previously been deployed in Qur'anic studies as a unified analytic framework, although Qā'imīniyā (1390/2011, 1393/2014) and Pakatchi have integrated Izutsu and conceptual-metaphor theory, and Persian-language scalar-semantic work on Qur'anic vocabulary may exist in venues we have not been able to canvass. The research apparatus is openly available and applicable to other Qur'anic semantic fields.

Descriptively, we have produced an exhaustive, reproducible concordance of 312 attestations across the fourteen core spectrum roots, with corresponding distributional and network-analytic outputs. The CSV outputs — including FDR-adjusted binomial tests, marginal-preserving permutation null p-values, and bootstrap 95% CIs on every centrality estimate — provide a foundation on which further philological and computational work can build.

Theoretically, we have formulated and empirically defended the Qur'anic Action-Intensity Continuum Hypothesis: the claim that fourteen core lexemes of the Qur'anic anger-spectrum field operate as a single graded continuum structured by intensity of action and partitioned into six phenomenologically meaningful stages. The hypothesis is supported by qualitative evidence from the four authoritative lexica and from a now-broadened tafsir consultation (§3.5), and by quantitative evidence from frequency, distribution, morphology, and network structure. We have, in addition, identified distributional patterns that, to our knowledge, have not been previously reported as a coordinated set, while reporting their post-FDR evidential strength transparently: the exclusively-Medinan sakhaṭ survives Holm-Bonferroni correction at $\alpha = 0.05$ borderline (raw $p = 0.005$, adjusted $p = 0.05$) and is robust across baselines 0.70–0.78; the exclusively-Meccan asaf, the Meccan-tilted naqm, the hapax-Meccan ḥard, and the Medinan-tilted karh are reported as contextual observations warranting follow-up rather than as confirmed findings. The general phenomenon of discursive-contextual lexical specialisation of which we hypothesise these are instances is proposed as a research target rather than as a settled result. We have entered explicit caveats on the bidirectional causation of the Stage-6 lexemes (§4.8.5), grounded in a $\kappa = 0.79$-validated tafsir-coding audit, and argued that Q. 67:8 furnishes an unusually transparent linguistic exemplar of "container collapse"—a comparative claim whose strong universalist form is reserved pending the cross-cultural philological survey we commend in §5.5 and §6.3.

6.2. Answers to the research questions

RQ1. The semantic field is a fourteen-node network organised around the bridge node baghy and the stage-internal hubs ḥuzn and ghaḍab, with several specialised peripheral nodes (sakhaṭ, asaf, ḥard, ʿutuww) exhibiting marked distributional asymmetries.

RQ2. The fourteen nodes order on a six-stage continuum—pre-anger displeasure (Stage 1), inner pressure & contraction (Stage 2), evaluative aversion (Stage 3), active anger (Stage 4), compressed/explosive rage (Stage 5), behavioural outcomes with caveats (Stage 6)—organised by the composite dimension of intensity of action (locus, intensity, scope of consequence).

RQ3. The empirical distribution is consistent with (though not by itself confirmatory of) the continuum: the asymptotic chi-squared rejects the null of six-stage uniform distribution ($\chi^2(5) = 227.15$; large effect size at Cohen's $w = 0.85$), but a marginal-preserving permutation null on the same statistic yields $p = 0.24$, indicating that the asymptotic p-value over-states the evidence on this sparse, marginally-imbalanced design; the substantive case for the six-stage architecture rests on the qualitative lexicographic, exegetical, and metaphorical evidence rather than on the omnibus p-value alone. The comparison between Stages 1–5 and Stage 6 fails to reject equal split ($\chi^2(1) = 1.55$, Cohen's w = 0.07, trivial effect); we read this as consistent with — but not as positive evidence for — the bidirectional-causation reading of Stage 6, with the substantive support coming from the $\kappa = 0.79$-validated tafsir-coding audit (§4.8.5). The network exhibits a bridging-role profile at baghy with wide bootstrap confidence intervals reported transparently. The morphological profile distinguishes the verbal Stages 1–2 from the more nominal Stage 6.

RQ4. The continuum is structurally homologous with Gross's process model, Plutchik's wheel, Spielberger's STAXI, and Lazarus–Scherer appraisal theory in its psychological component, but extends beyond all of them by integrating the post-failure structural-moral consequences of unmanaged emotion (Stage 6) — while qualifying that integration with the bidirectional-causation caveat of §4.8.5.

RQ5. Conditional practical implications include candidate distinct-equivalent translations and position-marking conventions (subject to a 30-verse stratified follow-up evaluation, §5.6) and three Islamic-psychology applications (early-stage prevention, kaẓm-grounded crisis intervention, tawba-grounded rehabilitation). All practical implications are framed as hypotheses for empirical follow-up, not as settled prescriptions.

6.3. Limitations and future work

Five limitations bound the study: (i) surface-lexical scope — discourse-analytic close-reading of the spectrum's narrative deployment across suras is left for future work; (ii) the broadened nine-tafsir engagement is still selective, with deeper Sufi (al-Qushayrī, Ibn ʿArabī's school) and Persian-language commentary needing fuller integration; (iii) the cognitive-linguistic apparatus is Anglophone-dominated — engagement with classical Arabic bayān theory (al-Jurjānī's Dalāʾil al-Iʿjāz) would supply an indigenous substrate; (iv) the ḥadīth corpus has not been integrated; (v) the §5.7 Islamic-psychology implications await operationalisation in pilot studies of kaẓm-grounded emotion-regulation training. A natural follow-up is the AraBERT/CAMeLBERT contextual-embedding approach to the same fourteen-root inventory (testing stage-recoverability from distributional structure), deferred here to maintain the philological-statistical centre of gravity.

6.4. Closing remark

The Qur'anic lexicon encodes a sophisticated theory of the anger spectrum — not as explicit theoretical formulation but as the architectural design of its semantic surface. Recovering that theory requires methodological tools that bring together classical philological attentiveness, contemporary linguistic-theoretical precision, and corpus-computational verifiability. We hope the present study, alongside Izutsu, Qāʾimīniyā, Pakatchi, Narimani et al., and Qarizadeh et al., advances this triple integration. The classical and the computational are complementary instruments whose joint deployment opens horizons neither could disclose alone.

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Competing interests

The authors declare no competing financial or non-financial interests.

Ethics statement

This study uses only publicly-released text-corpus data (the Quranic Arabic Corpus, GPL-licensed; the Tanzil Quran text, CC-BY-ND 3.0). No human or animal subjects were involved at any stage. As a corpus-only philological-computational analysis, the work is exempt from human-subjects review under the standards of Urmia University and of comparable institutions.

Author contributions

K.J. (corresponding author) conceived the project, developed the theoretical framework, conducted the lexicographic and tafsir-tradition analysis, drafted the manuscript, and is responsible for the final form of the argument. A.J. contributed to the theoretical framework, the design of the computational pipeline, and editorial revision. Both authors approved the final manuscript and are jointly accountable for its content.

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Data and Code Availability

All code, data, concordance CSV outputs, and figure-generation scripts are released open-source at:

github.com/ali-kin4/quranic-emotion-spectrum

The repository contains: (i) the Python pipeline under MIT License; (ii) the Quranic Arabic Corpus (Dukes 2011) under its original GPL; (iii) the complete CSV concordance for the fourteen core spectrum roots (312 attestations), distributional summaries, statistical tests ($\chi^2(5) = 227.15$; $\chi^2(1) = 1.55$), network-centrality outputs, and umbrella-cooccurrence tables; (iv) PDF and PNG versions of the seven figures published with this paper; (v) full text of the Persian and English manuscripts; (vi) the SHA-256 hash of the QAC v0.4 source file (`data/quran/qac_morphology.sha256`) and the manual-validation audit (`data/concordance/validation_audi`
The complete analysis can be regenerated end-to-end by:

```
pip install -r analysis/requirements.txt
python analysis/extract_concordance.py
```

```
python analysis/advanced_metrics.py
python analysis/visualize_spectrum.py
python analysis/visualize_advanced.py
```

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